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**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

**Academic Year: 2024- 2025 (Even Semester)**

## Active Learning Best practices: Think Pair Share - Cooperative Learning Technique

| | |
|---|---|
| **Degree/Branch** | **: B.Tech. / Artificial Intelligence and Data Science** |
| **Year/Semester & Section** | **: I Year / II Semester & "A" Section** |
| **Course Code & Title** | **: AD3251 – Data Structures Design** |
| **Name of the Faculty-Member** | **: Mrs.B.SankaraLakshmi, AP/AI&DS** |

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| | |
|---|---|
| **Theme of Discussion** | **: Binary Search Tree** |
| **Topics Covered** | **: Unit - IV** |
| **Date and Time** | **: 03.05.2025 & 11.35 PM to 12.20 PM** |

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|---|---|
| **Course Outcome** | **: CO4** |
| **Topic Learning Outcome** | **: TLO 35** |
| **Activity Chosen** | **: Think Pair Share** |

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**Active Learning Best practices: Think Pair Share - Cooperative Learning Technique**

**Topic** **: Binary Search Trees**

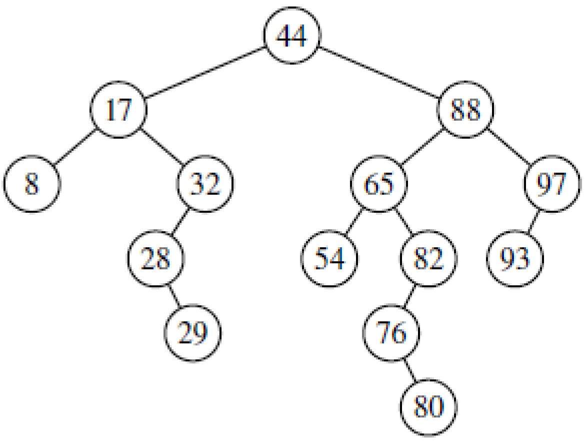


Figure 11.1: A binary search tree with integer keys.

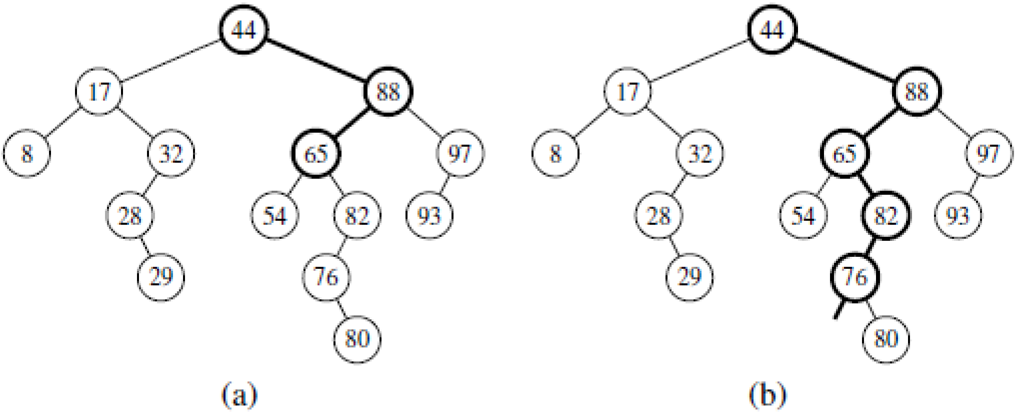


**Figure 11.2:** (a) A successful search for key 65 in a binary search tree; (b) an unsuccessful search for key 68 that terminates because there is no subtree to the left of the key 76.

**Justification:**

As part of the Think-Pair-Share method of collaborative learning, students are encouraged to think independently and share their insights with their peers.

- **Step:1 (5 Minutes)**

  The group of students listens attentively to the instructor's question.

- **Step: 2 (10 Minutes)**

  After reflecting for a while, each student writes down his or her answers.

- **Step: 3 (15 Minutes)**

  The students engage in a discussion with their peers to share and refine their responses.

- **Step: 4 (15 Minutes)**

  The teacher invites selected students to share their thoughts and ideas with the entire class.


**Time allotted for the activity: 45 Minutes**

- **Details of the Implementation:**

Think-Pair-Share, a collaborative active learning strategy, was conducted for the I AI & DS – ‘A’ section students. During this activity, students actively engaged with a question posed by the instructor, following the steps of individual thinking, peer discussion, and group sharing.

**Think (T)** : Students individually reflect on the given topic Binary Search Tree and write down their responses.

**Pair (P)** : Each student pairs up with their peers or joins groups to discuss the properties and applications of Binary Search Tree.

**Share (S)** : Students engaged in discussions with their peers and extended the conversation to a whole-class discussion. Alagu Siva Sambath A and Manosri E from the I AI & DS ‘A’ section shared their insights with the entire class.

- **CO–PO/PSO Mapping:**

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO9 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO4 | 3 | 3 | 3 | 3 | 1 | 3 | 2 | 2 | 3 | 2 | 3 |

**( 3 – High 2 – Medium 1 – Low)**

- **PO/PSO Mapped:**

| Innovative Practice | PO1 | PO2 | PO3 | PO4 | PO5 | PO9 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Justification for Correlation | Apply the knowledge of Tree Structures to the solution of Complex Engineering Problems | Identify and Formulate the structure of BST to solve the problems of searching, indexing and sorting | Design solutions for complex engineering problems using BST structure | Use research based knowledge including design of experiments and analysis to provide valid conclusions | Select and Apply appropriate techniques to complex engineering activities | Function effectively as an individual and as a member in developing application | Demonstrate knowledge and understanding of BST structure and apply this to ones own work to manage projects | Recognize the need and have the preparation and ability to engage in independent and life - long learning in the brodest context of BST structures | To apply analytic technologies of tree structures for solving business and engineering problems | To create and apply the techniques of tree structures in the domain of HealthCare, Education etc., | Apply tree structures to foresee research findings and provide the solutions |

## Glimpses:


GPS Map Camera
Rajapalayam, Tamil Nadu, India
Main Building, Vadakku Venganallur, Rajapalayam, Tamil Nadu 626125, India
Lat 9.482635° Long 77.514277°
03/05/2025 12:03 PM GMT +05:30
Google


GPS Map Camera
Rajapalayam, Tamil Nadu, India
Main Building, Vadakku Venganallur, Rajapalayam, Tamil Nadu 626125, India
Lat 9.482677° Long 77.514139°
03/05/2025 11:57 AM GMT +05:30
Google


❖ **Reflective Critique:**

✓ **Feedback from students and stakeholders:**

- Students felt they had adequate time to think critically during the activity.
- They also appreciated the opportunity it provided to collaborate in groups.

✓ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students gain better clarity on the topic by discussing and sharing their views with peers.

✓ **Challenges faced in implementation:**

- One representative from each group is selected to present their answers to the class.

❖ **References:**

https://www.readingrockets.org/strategies/think-pair-share
https://www.kent.edu/ctl/think-pair-share
https://www.readingrockets.org/classroom/classroom-strategies/think-pair-share

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**Feedback Questionnaire:**

1. How effectively was the Innovative Practice Plan implemented?
   **Excellent** **Good** **Satisfactory**

2. Did the plan align with your expectations?
   **Yes** **No**

3. How well did you understand the Objectives and Purpose of the Innovative Practice Plan?
   **1 2 3 4 5**

4. Were the necessary resources and support provided for successful implementation?
   **Yes** **No**

5. Did you notice any specific benefits or changes resulting from this plan?
   **Yes** **No**

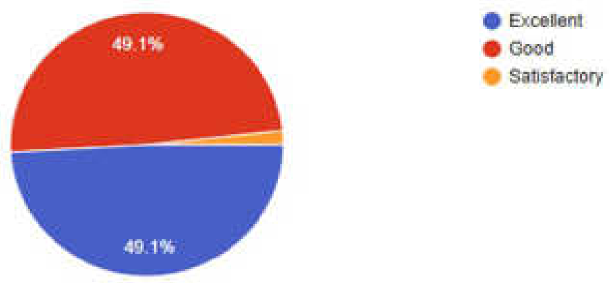
6. How were you engaged throughout the practice plan?
   **1 2 3 4 5**

7. Did the plan provide opportunities for Active Participation and Collaboration?
   **Yes** **No**

## Feedback Analysis:

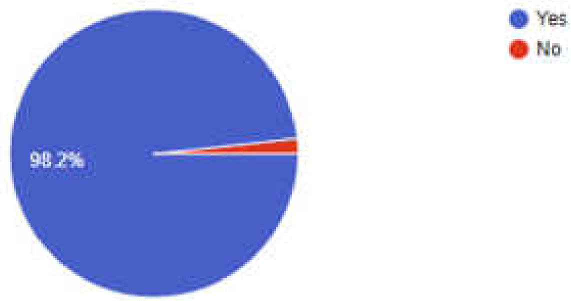
1. How effectively was the Innovative Practice Plan implemented?

57 responses



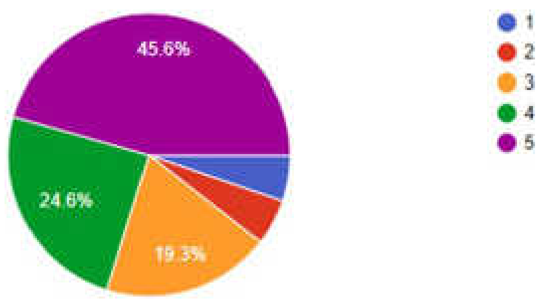

2. Did the plan align with your expectations?

57 responses



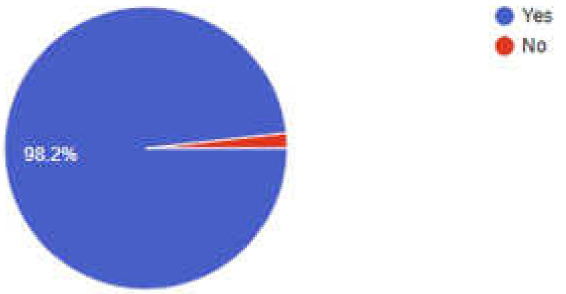

3. How well did u understand the objectives and purpose of the innovative practice plan?

57 responses



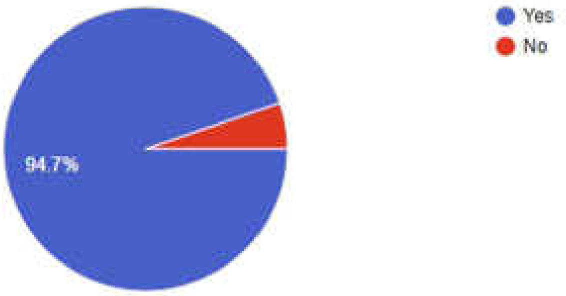
4. Were the necessary resources and support provided for successful implementation?

57 responses



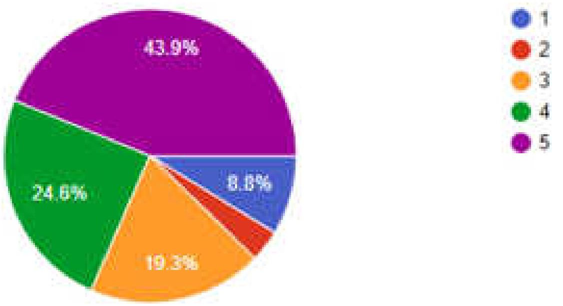

5. Did you notice any specific benefits or changes resulting from the plan?

57 responses



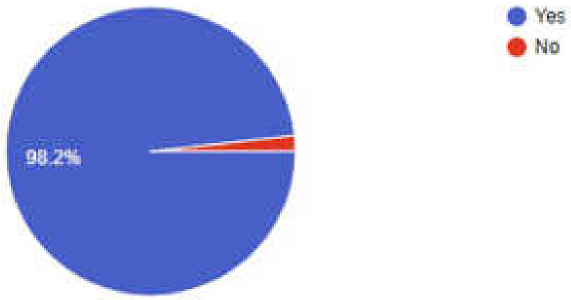

6. How engaged were you through out the practice plan?

57 responses



7. Did the plan provide opportunities for active participation and collaboration?

57 responses




**Signature of Faculty Member** **HOD**